

GAMIFIY LEARNING GAMIFICATION IN UX DESIGN

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GAMIFY LEARNING

GAMIFICATION IN UX DESIGN

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1. INTRODUCTION

In recent years, there has been a significant rise in the use of mobile applications and games due to the rapid advancement and increasing accessibility of digital technologies. Gradually, these technologies have become a part of daily lives and because of that, markets have developed further, especially in entertainment, content creation, and education. From these categories, a hybrid is developing a growing attention: edutainment.

The word “edutainment” comes from the combination of the two words education and entertainment. The goal is to increase interest for learners by using the elements of storytelling.⁰¹ Often, it gets confused with gamification, but there is a key difference between them: Gamification uses the game mechanics, by introducing score systems or creating a competitive environment, for instance. Whilst the goal in education is to motivate active participants and to help committing the material to memory.⁰² Games on the other hand have clear rules, goals, feedback etc. Its purpose is mainly entertainment.⁰³ Serious games are games but for educational purposes.⁰⁴

Lately, language apps have gained popularity through the years by marketing the apps in social media, for example Duolingo with its marketing strategies.

However, does gamification help with second language acquisition?

This analysis will reflect on gamification elements in different apps from the category language learning, specifically Duolingo, Babbel, LingoDeer, and Pingo AI and its user experience design with its AI features. It will examine its effectiveness and its support of learning outcomes. And it will critically examine gamification as a design element. It will also draw on motivation theory, flow theory and cognitive learning theory and examines how these frameworks use within the app’s design.

01 (Stangl, n. d.).

02 (Kurt, 2023).

03 (Jacob, 2023).

04 (*Serious games - spielerisch Ernste Inhalte vermitteln*, n.d.).

The German and French courses are going to be examined, as German is my native language and French is not, for comparison between other courses. On this analysis, the Premium Versions of the apps, such as Duolingo MAX, Babbel Premium, LingoDeer Premium and Pingo AI Premium, are going to be studied.

2. THEORETICAL BACKGROUND

2.1 The Western History of Language Teaching

One of the first documented teaching method was the Grammar Translation Method or the Prussian method. This method dominated from the 1840s until the 1940s. Some parts of the world still use this method. The method was documented and described by Barnas Sears in 1845, an American teacher. The characteristics of this method are, that the grammar rules are being taught first and then applying the knowledge into accurate translating. The native language is used as instruction.

Yet this method was heavily criticized by many, and it resulted in students having a distaste of the target language. With time, opposition grew to this method. Phonetic methods were more preferred. Claude Marcel, a language teacher, emphasized child language learning as a model. He proposed that reading should be learned before learning other skills.

From François Gouin observations of children acquiring language, he developed teaching a foreign language from children's approach of language. He believed, language is based on context depending on events. He developed the Gouin series and it was popular in his time (Img 01). Elements from this method became a part of other teaching methods like Situational Language Teaching and Total Physical Response. Yet Gouins and Marcells teaching methods didn't receive much support or attention.

With time, the emergence of the Reform Movement in language teaching started. More educators recognized the importance of speaking proficiency than reading comprehension, grammar and literary appreciation. In 1886, the International Phonetic Association was founded. One of the first goals was to improve teaching of modern languages.

Over some time, natural language learning principles became the foundation for the Direct Method. The method was introduced in France and Germany after the success of the commercial schools from Sauveur and Maximilian Berlitz. The following principles were, that daily vocabulary was taught in an oral communication. Specific words were taught by demonstrating or showing images and objects. Abstract vocabulary was taught in associating ideas.

I walk toward the door.	I walk.
I draw near to the door.	I draw near.
I draw nearer to the door.	I draw nearer.
I get to the door.	I get to.
I stop at the door.	I stop.
I stretch out my arm.	I stretch out.
I take hold of the handle.	I take hold.
I turn the handle.	I turn.
I open the door.	I open.
I pull the door.	I pull.
The door moves.	moves
The door turns on its hinges	turns
The door turns and turns.	turns
I open the door wide.	I open.
I let go of the handle.	I let go.

However, it was difficult to introduce the Direct Method in public secondary school as it didn't consider the practical reality of the classroom. It lacked the basics of applied linguistics, and sometimes it was easier to explain theory in the native language instead of the course. By the 1920s, the Direct Method had declined in European state schools.

Through further development, new teaching methods such as the Situational Language Teaching were introduced in the 1920s and 1930s. This theory encourages students to speak and listen to the course language, and accuracy was highly regarded by teachers, especially using the drilling method.

Around the 1950s to the 1980s, many methods emerged like the Audiolingual Method, Silent Way, the Natural Approach and the Total Physical Response.⁰⁵ Typical characteristics of Audiolingual Method are pattern drill, the fill-in-the-blank exercises, and learning by heart.⁰⁶ The principles of the Silent Way are, that teachers don't interfere, yet the students are encouraged to discover or create, use of objects and problem solving.⁰⁷ The Natural Approach emphasizes exposure by consuming the language by reading and listening before beginning to speak.⁰⁸ Total Physical Response uses coordination of speech and physical activity.⁰⁹

In the 1990s, Content Based Instruction, Task-Based Language Teaching, Competency-Based Instruction, Cooperative Learning, Whole Language Approach and Multiple Intelligences have been developed as teaching methods. Content Based Instruction has the philosophy of acquiring language through learning new information instead of the language itself.¹⁰ Task-Based Teaching orients in tasks, in which the students use language more authentically and the tasks itself focus on outcomes.¹¹ Competency-Based Teaching is a student-oriented approach in mastery of practical competences.¹² Cooperative Learning focuses on interaction between students by completing a task together.¹³

05 (Richards & Rodgers, 2001, p. 3-17).

06 (*Audiolinguale Methode*, 2025).

07 (Richards & Rodgers, 2001, p. 81-82).

08 (Richards & Rodgers, 2022).

09 (Richards & Rodgers, 2001, p. 73).

10 (*Content-based instruction / content and Language Integrated Learning. Methods of Language Teaching.*, n.d.).

11 (Van den Branden, n.d.).

12 (Bouchrika, 2026).

13 (Lewis, 2019), (Richards & Rodgers, 2001, p. 3-17).

2.2 Motivation Theory

Motivation is seen as the desire to accomplish a task, characterized by enthusiasm and determination to reach goals. Motivation can lead to positive or negative changes, depending on certain actions. There are many motivation theories, and the most well-known is the intrinsic and extrinsic theory of motivation.

The intrinsic theory explains that people are interested in certain behaviors or activities if they find them engaging. In the long term, it is more sustainable because of genuine interest, rather than rewards, it is the motivating factor. However, rewards can be relevant to intrinsic motivation, depending on individuals and context. The extrinsic theory sees that people engage in tasks for potential external rewards or pressures than the personal interest or satisfaction. It is characterized by external factors; for example, financial, academic or social rewards, to motivate.

The key concepts of Self-Determination Theory emphasize autonomy of free choice, competence of the individual's skill, and success and feeling of connection with people. The concept considers also both motivations within the context of these factors: Intrinsic motivation lets people engage in activities to improve or to achieve inner satisfaction. Extrinsic motivation is driven by social values, which drive other actions. It distinguishes from other theories by focusing on autonomy, choice and human relationships.¹⁴

2.3 The Definition and Young History of Gamification

Gamification is the use of game design elements in non-game content. This concept tries to trigger intrinsic and extrinsic motivation of the user. Intrinsic motivation is achieved by emotionally engaging the user to the activity and extrinsic motivation is driven by rewarding participants with grading, badges, achievements and financial in-game rewards. The goal of gamification is to increase motivation, boost material retention and motivate active participation.¹⁵

One of the earliest documented signs of applied gamification is the Boy Scout Movement in 1908. To enhance the efficiency of the organisation, members received rewards, for example special badges (Img. 02), by participating in activities.¹⁶

In 1984, “Game of Work” was published by Charles Conradt, in which he explored the theory of finding fun in work, similar to play. In his book, he insisted that the power of games would result in increased productivity and job satisfaction. Charles himself also gave consulting to firms about work organization.¹⁷

Around the beginning of the 80s, more loyalty programs in customer programs were introduced, especially in the airline industry. One big example is the American Airlines launching “AAdvantage”, in which the members received miles by looking how much the member has flown. With the miles, the member has the opportunity to redeem rewards out of it.¹⁸

Bret Terrill’s blogs documented the term “gamification” from 2008. He emphasised the potential for enhancing experience and motivation of its use.¹⁹ However, a British software engineer named Nick Pelling, claimed to coin the term in 2002.²⁰

In the 2010s a rise of gamification began to develop. Research around the theory of gamification began to be taken more seriously, especially after the official release of the book “Reality is Broken” by Jane McGonogal in 2011. It reflects, how gaming provi-

15 (Kurt, 2023).

16 (Scheidlinger, 1948, p. 740.), (Sharma & Sharma, 2023, p. 9).

17 (Krogue, 2012), (Sharma & Sharma, 2023, pp. 9-10).

18 (Shelper, 2025), (Sharma & Sharma, 2023, p. 10).

19 (Sanchez et al., 2019, p. 2), (Sharma & Sharma, 2023, p. 10).

20 (Pelling, 2011), (Sharma & Sharma, 2023, pp. 10-11).

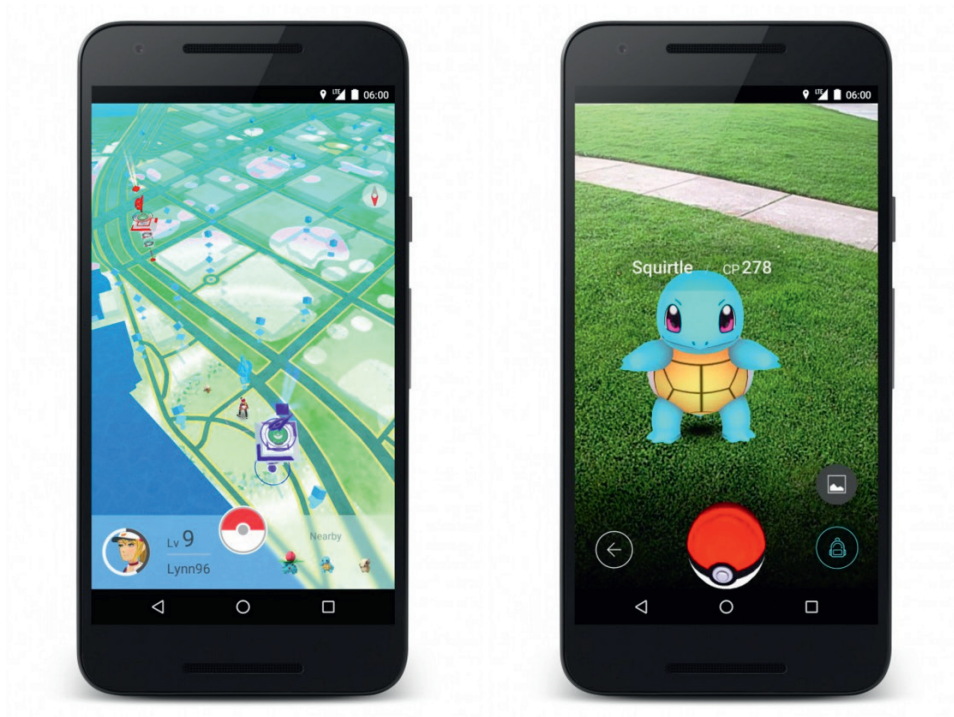


Image 02 (Up) A scout wearing his badge scarf, (Matsuzawa, 2019).
Image 03 (Down) Usage of the app Pokemon Go, (Chavez, 2016).

des people motivation, a sense of accomplishment and collaborative spirit, so that the daily life could be enrich.²¹ In 2011, a paper by Sebastian Deterding et. Al. was released about the modern meaning of gamification.²² Sebastian is a researcher focusing on UX, specifically about gamification.²³

Its peak reached after Pokémon Go's release in July 2016. The augmented reality game gained worldwide popularity by 800 million downloads. The gain of popularity was achieved not only by Pokémon's IP but also integrating gamified concepts into reality: Receiving rewards by going certain spots, social interaction with other users and AR features to immerse users into the game.²⁴ (Img. 03)

2.4 Flow Theory

The flow state is achieved by the individual's deep immersion in an activity, with enjoyment of the experience. The experience is so rewarding that they would engage just for the sake of it. Individuals could lose their sense of awareness of their surroundings through the flow state.

The balance between difficulty and skill is the most important part of the theory: if there is a right balance from a person, the flow state can be achieved. But if the person is skilled and the challenge is easy, they would feel boredom. Conversely, they would feel anxious. (Img. 04)

People with an autotelic personality, meaning they are self-motivated and goal-oriented, can experience flow. Flow can happen anytime, and the primary objective is to give pleasure and enjoyment. The difference between pleasure and enjoyment is important here: Pleasure comes from social or biological desires, like eating a meal. Enjoyment is more referred to as for a sense of accomplishment and novelty.²⁵

For example, Duolingo tries to achieve the flow state to the user by having the difficulty in the right place. Also, the app tries to find out, if the user already has previous knowledge of the language and would give the user a jump until the right level.

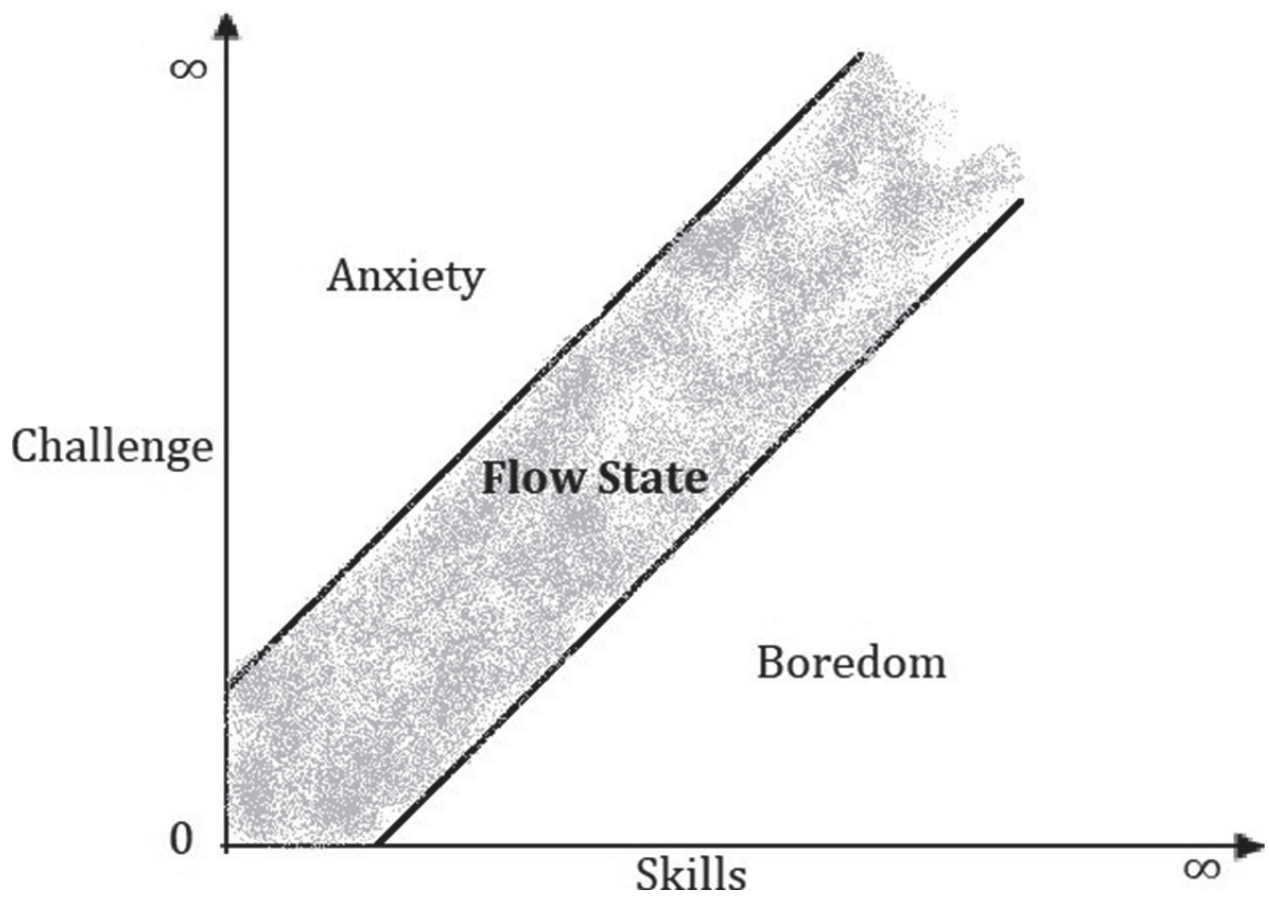
21 (Guldhe, 2023), (Sharma & Sharma, 2023).

22 (Deterding et al., 2011, pp. 1-3), (Sharma & Sharma, 2023, p. 11).

23 (Professor Sebastian Deterding professor of digital creativity, n.d.).

24 (Antara, 2025), (Sharma & Sharma, 2023, p. 11).

25 (Sharma & Joshi, 2025, pp. 297-303).

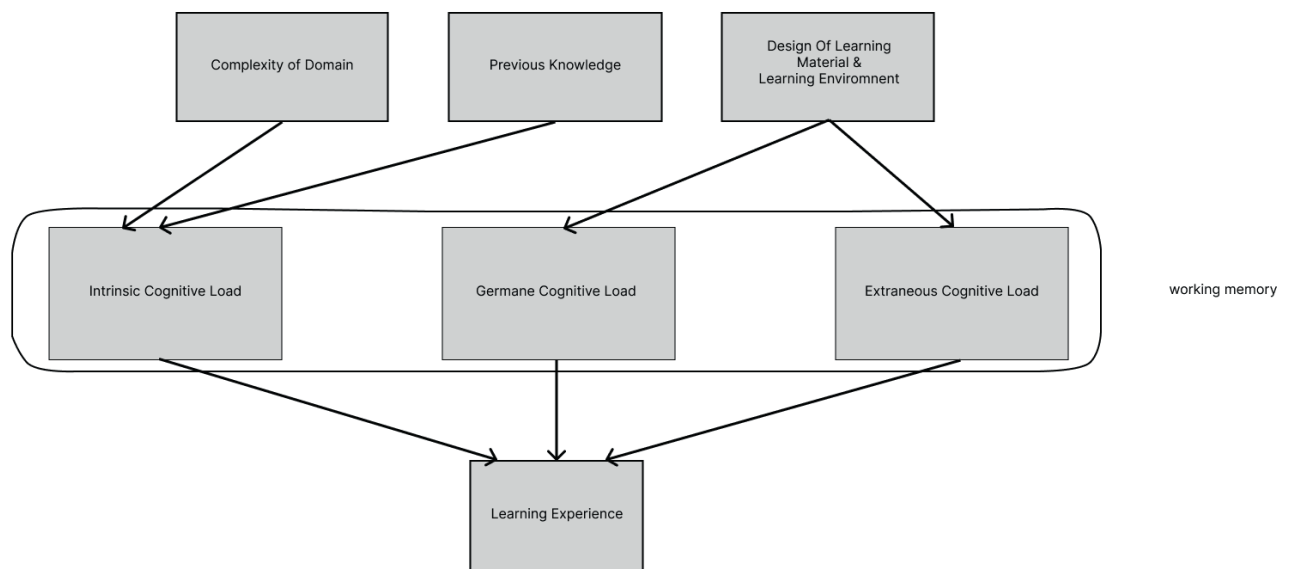


2.5 Cognitive Load Theory

Cognitive Load Theory focuses on the role of working memory in learning. Developed by John Sweller in 1988, it describes how the learning process depends on the brain's limited capacity. There are three kinds of cognitive load:

- Intrinsic Cognitive Load describes the dependence on the complexity of the learning material. Understanding complex, abstract contexts is more challenging than rote learning, e.g. learning vocabulary by heart is easier than understanding grammar. Depending on the learning material, some difficulties cannot be changed, but conveying parts of the concept can be designed for an easy learning experience.
- Extraneous Cognitive Load depends on the presentation or design of the information. It should be kept to a minimum so as not to overload the individual.
- Germane Cognitive Load looks at the learning process itself and builds on learning a new concept or linking it with prior knowledge. At the end, the formation of a mental pattern is built by germane cognitive load in the longterm memory. (Img. 05)

With the knowledge of this theory, gamified learning experiences can be designed so as not to overload the user with conveying the language, having a minimal User Interface design and coursework design. For example, Babbel contributes more to the intrinsic cognitive load by making small tasks of learning pronunciation of single words or short sentences instead of a long text.²⁶



3. CASE STUDIES

In the case studies, the main User Experience (UX) structure, the gamification elements, and the limitations of apps are going to be analysed. At the end, the apps are going to be comparatively analysed by looking at the emotional and UX design, and the Ais.

3.1 Duolingo

Duolingo is a language app, and it is among the most popular apps according to the App Store charts in the category education.²⁷ In 2012, it was initially released as a website as a nonprofit by the two founders Luis von Ahn and Severin Hacker. As a final decision, they chose to be a for-profit company for long-term sustainability.²⁸

Their vision was to make language learning accessible based on the experiences from Von Ahn during his visit in Guatemala, by actions such as launching Duolingo English Test in 2014.

In 2023, they released the subscription model Duolingo MAX, in which the creators integrated advanced AI for features, e.g., Video Call and Roleplay.²⁸ In April 2025, the company decided to have an “AI-first” strategy, by integrating AI into their workflow.³⁰

3.1.1 Main UX Structure

Before beginning a lesson, the user is shown a field displaying the lessons and their progress. The start screen is designed in the style of a board game, with characters similar to game pieces (Img. 06). Continually, the mascot “Duo,” a green owl, accompanies the user on their learning journey by appearing on the main page, teaching lessons, and rewarding the user with in-game currency. Not only is Duo visible on the main page, but other recurring characters are visible on the app.

27 (Apple, n. d.).

28 (Wallingford, 2024).

29 (Duolingo, Inc. (DUOL): History, ownership, Mission, How It Works Makes Money, n. d.).

30 (Duolingo, 2025).

3.1.2 Gamification Elements

Before the user begins with their learning journey, the app or the mascot Duo asks, which proficiency level of the user's language is, from beginner to advanced. The app then offers the learner to evaluate their proficiency. Depending on the test results, the user can skip some lessons or start from the beginning.

The lessons are offered in packages, each dedicated to a topic. The more advanced the user is, the more complex the lessons become. In the lessons, the user is given various tasks: reading or saying sentences aloud, arranging words, filling out sentences, or translating.

During and after lessons, Duo is encouraging the user to keep going. At the end of each lesson, they say a catchphrase with astounding animations. At the same time the phone provides haptic feedback.

Besides the lessons, it contains a competitive feature, in which a group of users are in a league system: the rank changes, depending on how much experience the user gathers. It creates this environment that motivates users to gain more experience, so they could get into higher leagues or at least not get demoted.

One of the app's core features is the Streak feature, which encourages the user to practice on the app regularly. If the user completes a lesson, then they receive a streak. At midnight, the streak resets, unless the user has an ice streak in their inventory, in which case the streak continues as normal. After completing a lesson, Duo asks the user to set a streak goal. However, if the user does not take the lesson of the day, Duo will be disappointed. (Img. 07) Depending on how many consecutive streak days the user reaches, the rewards become more attractive.

During the day, the app notifies the user several times. The app not only reminds them through notifications (Img. 08), but the app also provides features such as putting a widget on the home screen for daily lesson reminders.

Case Studies

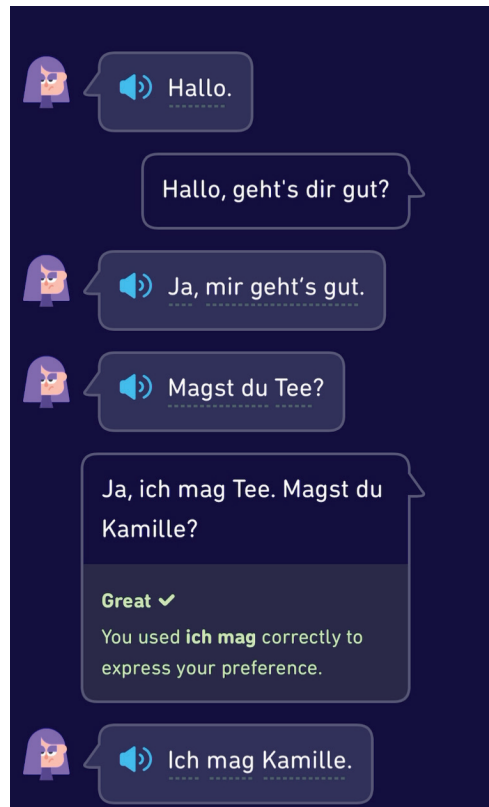
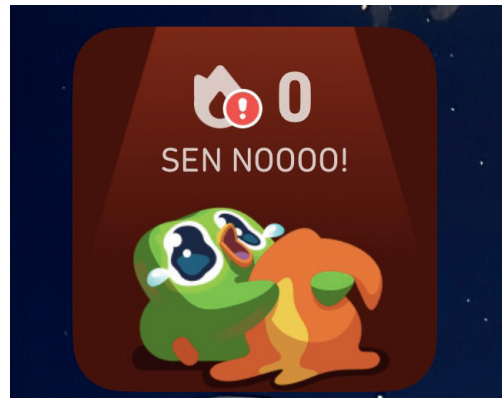
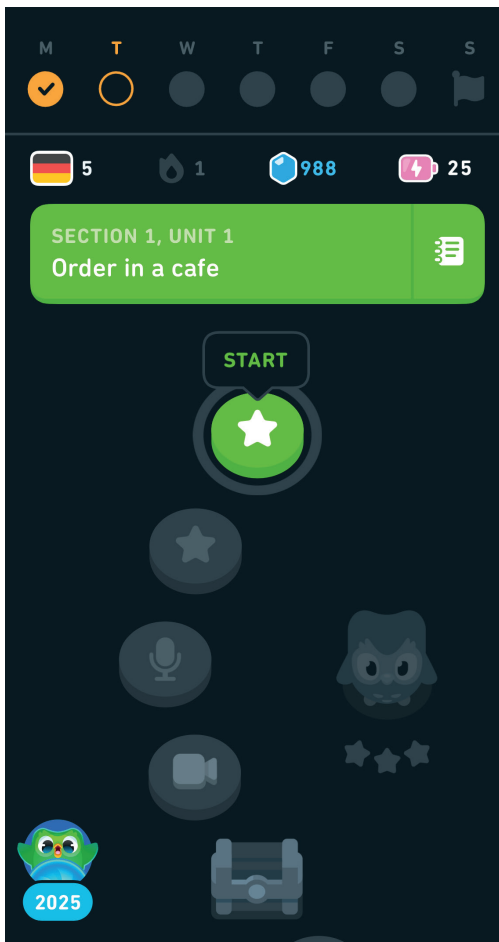


Image 06 (top left) Start page of Duolingo, (Duolingo, n.d.).
Image 07 (top right) If the consecutive streak is broken, (Duolingo, n.d.).
Image 08 (down left) Timer until the Streak is broken, (Duolingo, n.d.).
Image 09 (down right) A conversation log with Lily, (Duolingo, n.d.).

3.1.3 AI Features (Video Call and Lessons)

One feature of this app is the video call with the character Lily. Characteristically, she is blunt, but also friendly. The conversation does not go into the deep level but remains on the small talk area. Notably, the AI does not hallucinate, but it appears to be pre-structured. Hallucination of AI is, when the AI does not recognize the pattern of the input and creates outputs, that are nonsensical.³¹

After the call, the user can look up the conversation log. The logs also have some notes on how well the user talked and if they used the right vocabulary, which is helpful to improve in speech.

During the lessons, the user has also the opportunity to talk with an anthropomorphic bear named Falstaff. He is trying to be helpful during the lessons and switches from the native to the course language, so he can teach the user wording. He is careful with the user and tries to convey the language as simple as possible. The conversations also tend to be pre-structured.

3.1.4 Limitations and Flaws

During the usage, there are a lot of features like the lessons itself, the AI features, role play, podcasts and social media feed. Some of these features are hidden and are often forgotten to be used. For example, the roleplay feature is hidden.

Unfortunately, the AI does not try to help the user during the conversations. It will only answer on the course language and doesn't try to correct the user, except at the end of the lesson. (Img. 09) And the conversations feel less spontaneous and not longer than two minutes. Also, the conversations are only in the course language. There is no possibility of switching the native language during conversations, for example, asking questions about grammar, words etc.³²

31 (What are AI hallucinations?, n.d.).

32 (Duolingo, n.d.).

3.2 Babbel

The Babbel company was founded in Germany by Markus Witte, Thomas Holl, Toine Diepstraten, and Lorenz Heine. Their goal was also to make language learning more accessible and focus on real-life communication. At first, they began giving online courses through a web application. In 2013, their application for smartphones was released, which gained popularity.³³ At the end of 2023, Babbel introduced “daily conversations”, in which the AI only evaluated the pronunciation of the user. In 2024, they released the beta version for practicing speaking with an AI language model.³⁴

3.2.1 Main UX Elements

Before beginning their lesson, Babbel asks the user, which language they want to learn and why. Then, the user lands on the Start page, which is the Learn page (Img. 10). The lessons come with packs, except they focus more on the formal way by telling the user, which language proficiency it is made for.

Then the user has the possibility to learn the language, practice with digitalized cards, hear podcasts or conversations or talk with the Babbel AI each dedicated to a page. The user has also the ability to look at their progress and how many times they learned.

3.2.2 Gamification Elements

In the lessons, grammar is prioritised more than learning vocabulary. In the first German lesson, users learn how to use articles in the appropriate way (Img. 11). Vocabulary is also conveyed, but not as important.

The lessons are packed in units. Each unit has a theme, but each lesson is built up from learning vocabulary, pronunciation and grammar. Some of the tasks are very basic, for example speaking out the words. And in some tasks, the app explains the theory behind it and tests the user if they understood by giving tasks. Also, during the lesson, the app itself encourages the user to keep going with subtle glitter animations and positive comments to the user. (Img. 12)

33 (What is brief history of Babbel Company?, 2026), (Babbel, 2013).
34 (Jurran, 2024).

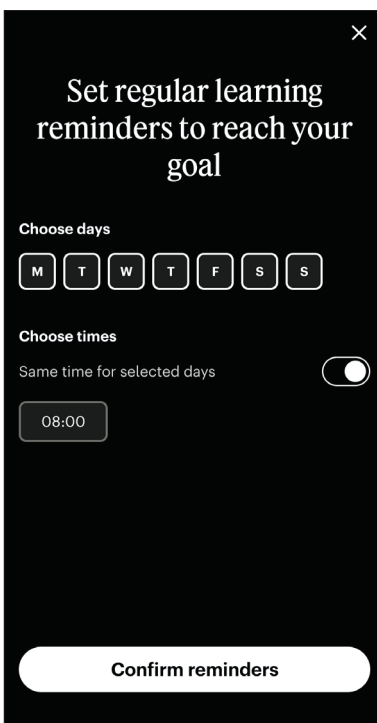
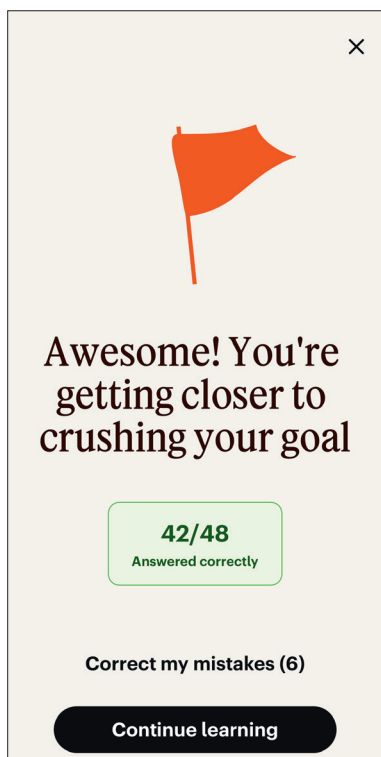
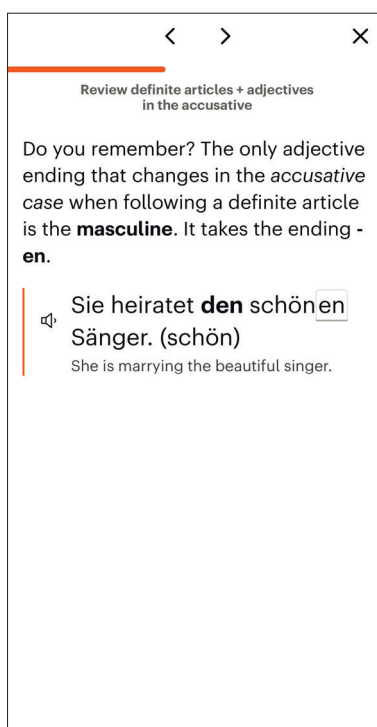
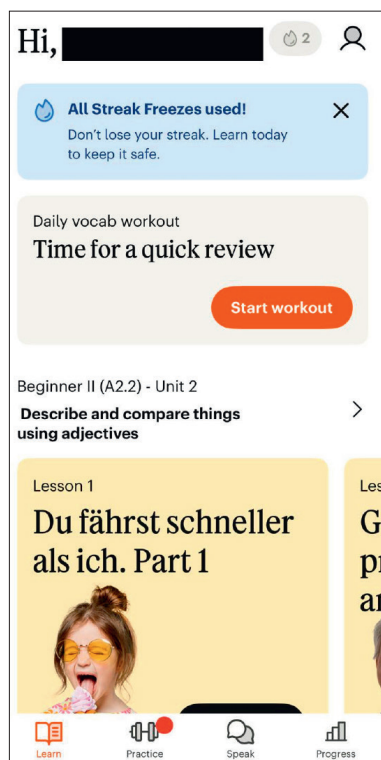


Image 10 (top left)
Image 11 (top right)
Image 12 (down left)
Image 13 (down right)

Learn page of Babel, (Babel GmbH, n.d.),
Explanation of grammar in the lesson, (Babel GmbH, n.d.).
After completing the lesson, (Babel GmbH, n.d.).
Setting reminders in Babel, (Babel GmbH, n.d.).

Gamification elements are not that apparent, yet a streak system is built in. The app does suggest setting some goals, and it reminds the user to come back for a lesson. The user can even set, when they need to be reminded. (Img. 13)

3.2.3 AI Features (Babbel Speak)

As a side note, the AI feature is still in the beta version, so it is not fully developed yet. The AI is built by having “scripted” conversations. Nonetheless, the user has a set of goals to reach, for example they need to bring up a certain topic. Even so, the overall topic can be chosen before beginning a conversation with the AI. At the end of each conversation, it shows a conversation log similar to the Duo AI, other than without any correction notes, but during the conversation, the user can look up tips. The voice is sometimes male, female or ambiguous.

3.2.4 Limitations and Flaws

Information distribution can be sometimes confusing because finding certain lessons to review again is difficult, or certain features can be forgotten. For example, users can listen to some podcasts. Unfortunately, the feature is hidden. During AI conversations, it doesn’t feel very spontaneous, because the goals of the conversations are rigid.³⁵

3.3 LingoDeer

LingoDeer was launched in 2017 by the founder, a computer scientist and language teacher named Wang Zhulong, from Hong Kong. Before he launched LingoDeer, he released a successful Mandarin app named “ChineseSkill”.³⁶ The app was specifically designed to teach Asian languages, yet today, they teach Western languages.³⁷

3.3.1 Main UX Elements

Before the user can utilise the app, the app also asks which language they want to learn. There are four buttons in the menu bar: Learn, Preview, and Me. Users can either participate in the lessons on the Learn page or practice from their learned vocabulary. Visually, the Learn page looks like a board game. The fields represent the lesson packs, and they are intertwined together. (Img. 14) LingoDeer does not have any AI features in the app.

3.3.2 Gamification Elements

There are also the tasks of translating, writing dictated texts and correcting sentences. During the lessons, there are two deer characters who assign tasks, one looks feminine and the other masculine. The characters do not have much personality during the lessons. The reason is that these two are the same character, except their genders got changed, and they always seem to be only happy.

At the end of the lesson, the characters congratulate the user and give a summary of their learned vocabulary, how much time it took, and how many errors the user made. Sometimes, at the end, the app gives rewards like experiences, badges and gems, which is the in-game currency. (Img. 15)

A useful feature is that the user can look up tips during the lessons. It does not say directly what the solution is, other than reciting explanations given before.

The app notifies users to take a lesson, yet only once per day. The app has a streak system. If the user forgets to take a lesson for the day, there is a way to extend the streak by having streak shields. (Img. 16) Another way to extend the streak is to have the streak

³⁶ (Scott, n. d.).
³⁷ (Trushyna, 2025).

Case Studies

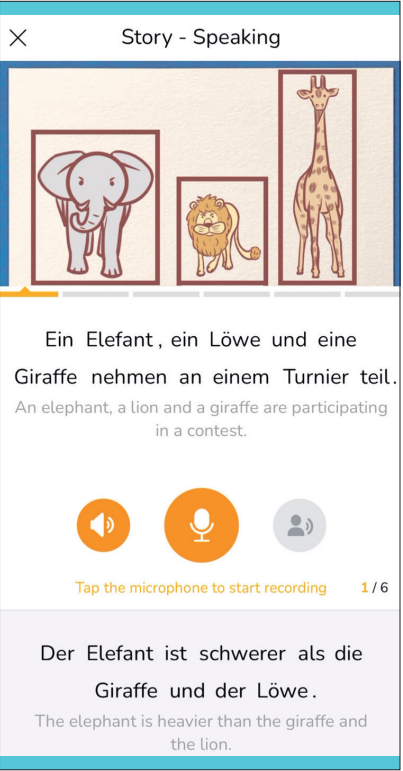
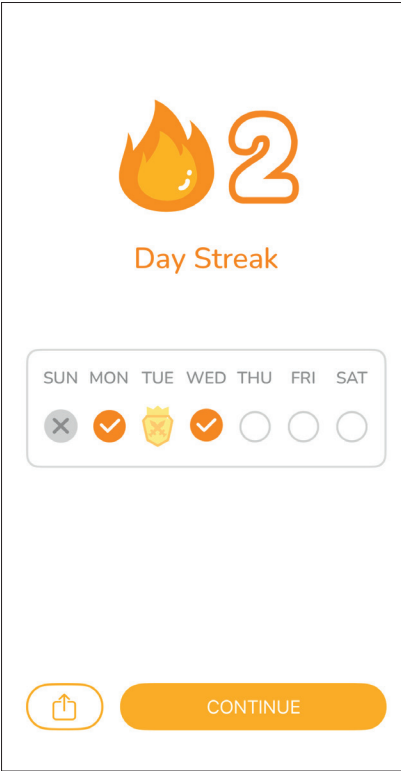
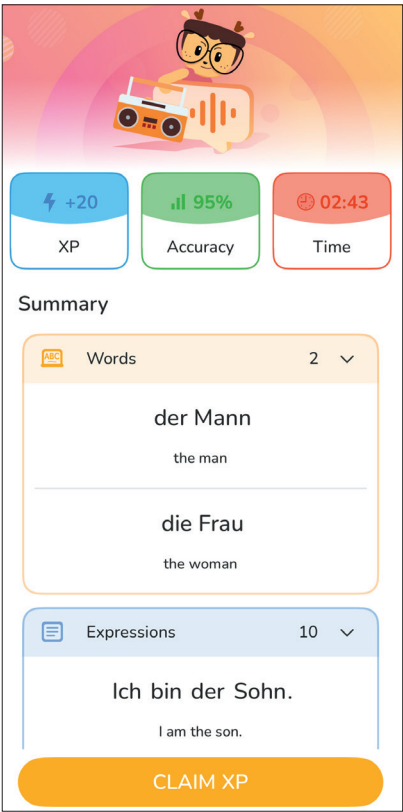


Image 14 (top left) Learn Page Of LingoDeer, (LingoDeer Co Ltd., n.d.).
Image 15 (top right) After Completing The Lesson, (LingoDeer Co Ltd., n.d.).
Image 16 (down left) Visualisation of the Streak Calendar, (LingoDeer Co Ltd., n.d.).
Image 17 (down right) During A Reading Comprehension Lesson, (LingoDeer Co Ltd., n.d.).

restorer. To get those items, the user needs to buy them with gems. In the lesson packs, it contains a story section, in which the user can practice more, in which they can read actively. To test their reading comprehension, the app asks a question during the lesson. It is supported with some images and illustrations to help the user to visualize the story more. (Img. 17)

There are two ranking systems. The first one is the league system, which ranks users by experiences gathered that week, and the other is the story reading ranking. The user gets either promoted to the higher league, or demoted to the lower one, depending on how many experiences the user has.

3.3.3 Limitations and Flaws

Voice recognition is not a feature of the app. It only records audio when the user is using the story lessons. For this reason, there is no real-time feedback on pronouncing the words the right way.³⁸

3.4 Pingo AI

The app was developed by the founders Michael Xing, a software developer, and Morrie Schonfeld, a published Chinese author.³⁹ The app was published in January 2025.⁴⁰ Right now, the user has access to learn 25 languages through the app.

3.4.1 Main UX Elements and AI Feature

The main feature of the app is talking with the AI. The AI is visualised by four dots and if it talks, the dots are elongated. The UI is simply visualized. The app uses a lot of rounded elements like on the buttons and typography. The AI is on the Speak page. The user can start a conversation any time and has the possibility to choose a topic or can go through a learning journey (Img. 18). During the conversation, a text appears when the AI speaks to the user. (Img. 19) The course conversations tend to be more one sided, but mostly because the AI teaches the user about new words and grammar. At the end of the talk, the user can look up the conversation log. The log can be also looked up in the history section from the Progress section.

The default voice of the AI is female, but the user can choose a male voice with a little change of personality.

38 (LingoDeer Co Ltd., n.d.).

39 (Pingo AI: *AI Companion that helps you learn languages*, n.d.).

40 (Xing, 2025).

Case Studies

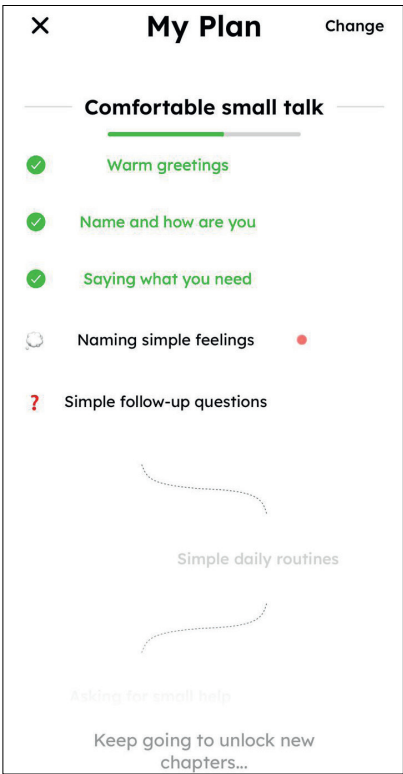
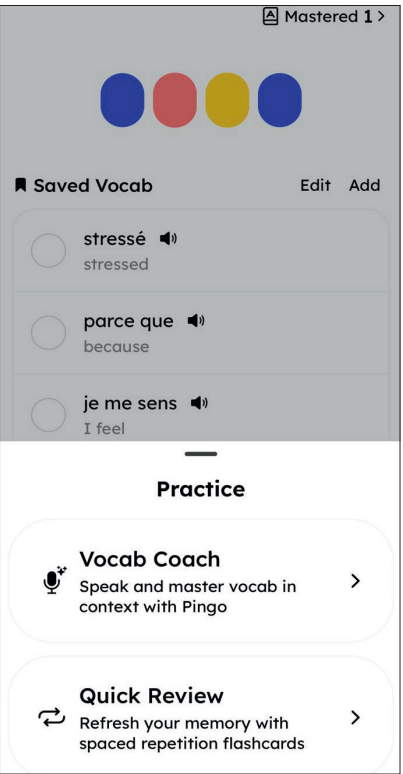
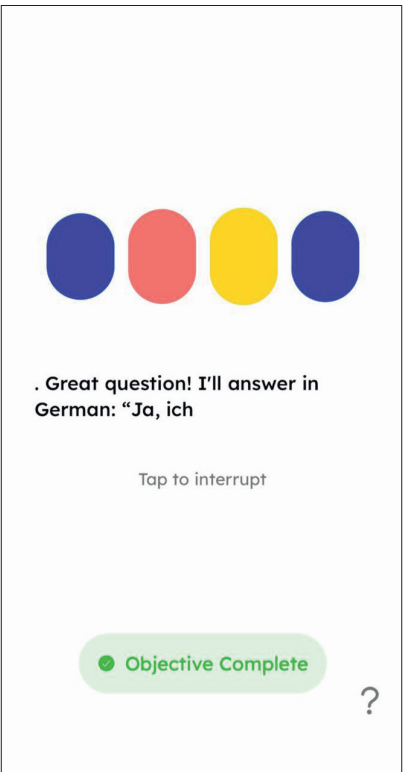
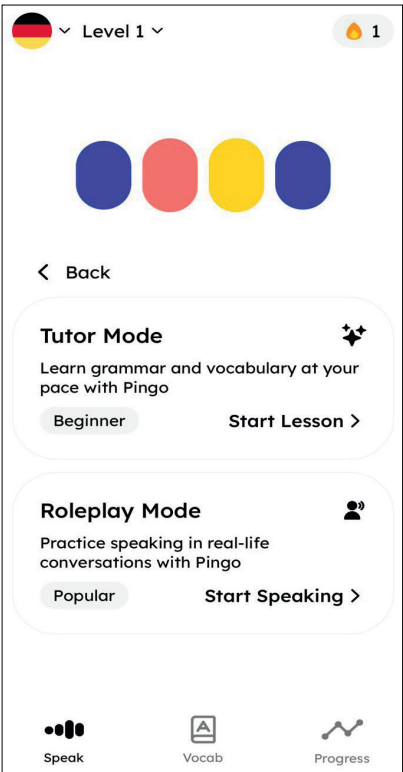


Image 18 (top left)
Image 19 (top right)
Image 20 (down left)
Image 21 (down right)

Modes of Pingo AI, (Pingo AI Inc., n.d.).
Talking with the AI, (Pingo AI Inc., n.d.).
Practice vocabulary, (Pingo AI Inc., n.d.).
Learning journey of the language course, (Pingo AI Inc., n.d.).

3.4.2 Gamification Elements

Through observation, there is not much of gamification elements on the app, except the basic streak system. Nonetheless, the user has the opportunity to look at their progress: Their estimated level of the language, from Beginner to Fluent, of the user, in which they count the number of learnt vocabularies.

Other gamification elements are collecting vocabulary from the conversations with the AI, or the user can generate the words. There are different tasks that the learner can use: Quick Review, in which they use a card system, and Vocab Coach, in which the AI practices the user with speech. There are two collections of vocabulary: the general collection and the mastered collection. If the user feels confident with the learnt word, they can sort it to mastered collection. (Img. 20)

Under the 4 dots, there is a button in which the user can see their learning journey. The learning journey is not pre-planned, instead the AI generates the courses itself (Img. 21), depending on what kind of answers the user gave on why they want to learn the language.

3.4.3 Limitations and Flaws

The language app mostly focuses on the oral part of the language learning, so there is no way to learn languages more formally when there is no or little written explanation on certain aspects. It mostly focuses on conversations and less on learning to write text. The AI also tends to be not that critical to the user. It only seems friendly, sometimes in a passive aggressive way, if the user made a mistake.⁴¹

4. COMPARATIVE ANALYSIS

4.1 UX and Emotional Design

Each app notifies users differently. On the one hand, Duolingo notifies the user more than once. Sometimes, the app tries to pressure the user to come back by having Duo die, because the user is not taking a course for several days. On the other hand, Babbel lets them customise when they need to be notified. Just LingoDeer and Pingo AI notify the user once or twice per day.

All apps have a friendly UX by using warm colours and round shapes. Duolingo and LingoDeer are also presenting mascot characters into their system.

Compared the apps overall, Duolingo has the most content with realized characters, even if it is the freemium version. The other apps not that much.

4.2 Gamification Elements

All apps have one thing in common: All of them have a streak system. In Duolingo, for example, the system is heavily gamified. This is because the app rewards users for achieving certain goals. The least gamified app is Babbel and Pingo AI. Both have a calendar to track the streak, despite the fact that they don't reward the user. LingoDeer does reward the user, but it does not give any clear goals.

Peculiarly, all apps have similar interfaces. In the lesson mode, all of them have a progress bar and similar visuals, if the given answer from the user is right or wrong. LingoDeer, especially, has an interface, like the progress bar during the lesson, and has a mascot.

The only exception is Pingo AI, in which the interface focuses more on the AI than the individual tasks of it.

4.3 The AIs

Duolingo and Pingo AI both have a female voice as the default option. Duolingo's AI only has the female voice, and Pingo AI is at least customizable. Babbel's AI changes time to time, depending on what kind of conversation. In Duolingo, it feels more natural to talk with the AI since there are animated visuals which come with the conversation. Babbel AI uses the conversation log

more actively, and it supports the learner about the conversation. It has a bit more gamified elements to it for giving the user goals. For example, one of the goals, the user needs to talk about how the AI is feeling. Still, the conversation feels more rigid than freely. Pingo AI, however, has more dynamism in the conversation, but it depends on the mode of the AI. In the roleplay section, it feels more like a person to talk to. In the learning section, however, it is more structured because the AI acts as a teacher. Babbel and Duolingo are more structured, but Pingo is more spontaneous, and users have the opportunity to customise the conversation by changing the language level, for example.

5. DOES GAMIFICATION SUPPORT LANGUAGE LEARNING?

During my research, I tried to arrange interviews with experts in gamification for language apps at various academic institutes. Fortunately, Dr. Zeynep Köylü from the Department of Languages and Literatures at the University of Basel agreed to be interviewed. Dr. Köylü is a lecturer in Second Language Acquisition and focuses on her research in computer-assisted language teaching.⁴²

Do language apps help with second language acquisition?
And does gamification support language learning?

If gamification is well implemented, it is possible to make learning fun. A key factor of gamification is that it tackles intrinsic and extrinsic motivation. Intrinsic motivation is the satisfaction of doing the activity itself, and extrinsic motivation is driven by rewarding participants with grades or badges. For example, LingoDeer gives the user badges and gems for completing lessons, which triggers extrinsic motivation.

However, pressure can create stress, burnout or annoyance to the user by push notifications, guilt-based messages and streak pressure, which could act counterintuitively to intrinsic motivation. For instance, Duolingo pressures the user daily, even with a timer, to come back and practice.

If the user develops emotional attachment, they could be more persistent during their learning journey. For example, Duolingo does it well by having characters around with some unique personalities. Also, the mascot Duo is also a well-rounded character who not only is happy if users take part in the lessons, but also gets furious if users do not complete their streak, or gets sad if users break their streak. The AI phone call is also a great feature in which the user would develop attachment to the characters even more. (Img. 22)

Yet, there is a danger of emotional distress from the user side because of the characters. The characters could guilt-trip the user into using the app, which would result in them not having fun while learning. For example, Dr. Köylü described an observation that she made with her child. They wanted to learn Korean, so they used Duolingo to learn. She only used the app on the weekends because of screen time restrictions. She could not do many

streaks, so the green owl would “die” (Img. 23); then she was not using the app daily.⁴³

Language apps offer the possibility of learning a language through more digestible content, so it is not overwhelming to start learning a new language. According to the Cognitive Load Theory, if the intrinsic cognitive load is used in the right way, the gamified software can create bite-sized content for the user to understand the language more. For example, Babbel makes short tasks about translating short sentences or words, instead of translating a whole text. (Img. 24)

Nevertheless, it also could make the learning experience more dreadful. Over time, it could lead to repetitiveness; if the tasks are very similar, it could demotivate the learner. The reason is that the tasks could become more predictable, e.g., Lingo-Deer has a difficulty putting diversity in the lessons. There are different tasks, still, the tasks themselves become predictive.

But language apps could potentially activate the “flow state” by using gamification. In theory, if the balance of skill and difficulty is reached, then the user could potentially learn the language effortlessly. For example, Duolingo tries to find out if the user knows any previous knowledge of the language, at the right level.

Unfortunately, other apps have failed to make personalised tasks. For that reason, the flow state could not be achieved, and learning would be difficult. For example, Pingo AI asks the user about their previous knowledge of the language, instead of giving them a little test to give them a specific level. If the user chooses the wrong level, it would be either too difficult or too easy.

Social elements could motivate users to practice more than normally they would. The reason is that it would influence extrinsic motivation through social comparison, e. g. leaderboards, leagues, or through the friends list. For example, Pokémon Go’s popularity was not only through the Pokémon, but also through social aspects of the game. The users have the opportunity to fight with their Pokémons or the users meet at PokéStops. It is also similar to Duolingo, which has a league system where users compete with each other to level up to the next best league. (Img. 25)

Does Gamification Support Language Learning?

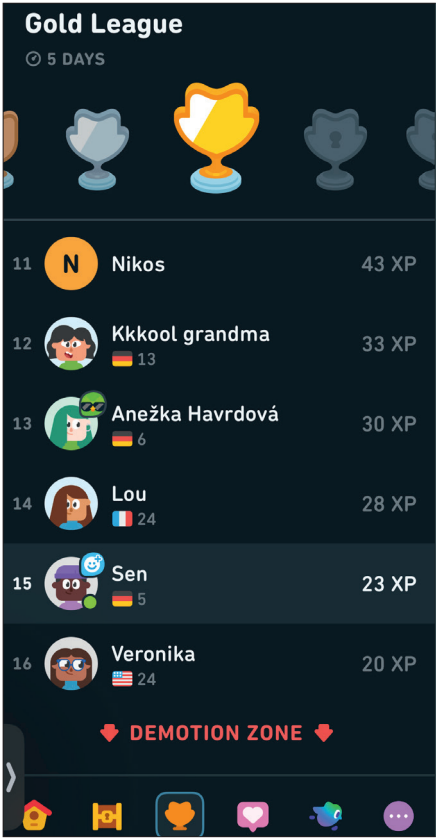
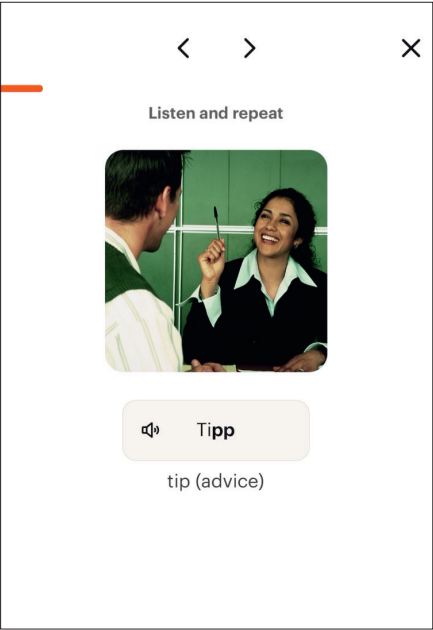


Image 22	(top left)	Lily Conversation, (Duolingo, n.d.).
Image 23	(top right)	Duo „Dying“, If The User Ignores The App for a long time, (Duolingo, n.d.).
Image 24	(down left)	A Small Oral Task in Babbel, (Babbel GmbH, n.d.).
Image 25	(down right)	League Ranking in Duolingo, (Duolingo, n.d.).

Yet, there is a risk of oversimplifying language skills through gamification aspects, because users learn through the “game”. The gamification aspect could simplify the language way too much, so some critical grammar could be omitted or forgotten by the user, such as Pingo AI not explaining enough about the grammar of the language.

Unfavourably, language apps could “stereotype” a language by using words that a native speaker would never say. Especially with the use of AI, it could make the stereotyping of a language a lot worse. Depending on how the AI is trained, it could talk in the way they are trained. For example, Pingo AI uses “wunderbar” a lot during the lessons, even though a native speaker would never use it often. The learner could also sound weird for the native speaker, or use it to mock people. Another example would be that the user also learns words that are “typical” to the origin of the language. For instance, in the French course of Duolingo, it teaches a lot more about bakeries, and in the German version, about alcoholic beverages.

According to Dr. Köylü, she sees the potential of gamification by learning the vocabulary, or in other words, the learners have better retention. From her research, there were three groups of language learners: teacher-learning method, online method, and blended method. The teacher method was the classical teacher–student dynamic, in which students had tasks in a typical classroom. In the online method, the students only learned in a learning management system (LMS)⁴⁴. And the blended method was the combination of both teacher and online methods. As a result, the teacher method performed the best, however, the LMS method had the best retention.⁴⁵

44 Learning Management System: A software for documenting, tracking, administrating, automating, and delivery of educational courses.

45 (Dr. Zeynep Köylü, 2026, Personal Communications).

6. CONCLUSION

But do language apps help with second language acquisition?
And does gamification help to learn the language?

It is remarkable how gamification has gained recognition in modern times. It has evolved from having no recognition to becoming a popular method of engaging in various activities, such as learning a new language. Thanks to psychological frameworks, such as motivation theory, the flow state, and the cognitive load theory, learning could be less exhausting. In the motivation theory, if the use of intrinsic and extrinsic motivation is used correctly and the difficulty and skills of the user is well balanced, it could lead to enjoyment in learning the language. However, learning should not be overwhelming, so the cognitive load would not overload the individual.

This thesis focused on four case studies: Duolingo, Babbel, LingoDeer and Pingo AI. Duolingo has the most gamification elements, but misuses emotional aspects with guilt-based mechanics. Pingo AI has a lower cognitive load, but the language course itself lacks structure.

The language apps have the potential to help with second language acquisition. With the gamification element, it can motivate learners to persist in their learning journey to acquire a new language, if the psychological mechanisms are well implemented, such as cognitive load theory and motivation theory. However, if certain aspects of gamification are misused, learning a new language would decrease greatly, and users would not come back to it. Especially if apps guilt-trip the user, or the tasks become repetitive. And apps are not the sole language acquisition tool: Users need to learn other ways to get the language, such as talking with other people, consuming media, listening to radio and reading books. Thanks to Dr. Zeynep Köylü's research in learning management system, learners of the digital courses have a better retention of the learning material than the teacher method. Language apps could be a part of our daily life, if we want to integrate it. Yet, more independent learners benefit the most.

I see the potential that gamification will be further develop with emerging technologies: Through AI, it could personalize the learning experience a lot more by making more individual tasks for the user. It could not only improve the commercial application but also it could help state education software, especially in today's world where keeping an attention span has gotten increasingly more difficult. I hope, gamification gets adapted into more industries such as healthcare and sports. Yet thanks to the popularity of the language apps, the possibility is looking brighter.

7. APPENDIX

7.1 Primary Source

Audiolinguale Methode. (2025, October 27). Retrieved March 27 2026, from ZUM Deutsch Lernen: https://deutsch-lernen.zum.de/wiki/Audiolinguale_Methode

Antara. (2025, March 12). *Pokémon go's success story: From launch to legacy*. Retrieved March 27 2026, from Analytics Insight: <https://www.analyticsinsight.net/games/pok%C3%A9mon-gos-success-story-from-launch-to-legacy>

Apple. (n.d.). *Die besten bildung-apps für iPhone*. Retrieved March 27 2026, from App Store: <https://apps.apple.com/ch/iphone/charts/6017?chart=top-free>

Arndt, H. (2022). *Digitale Spiele und fachliches Lernen: Band 1 (H. Arndt, Hrsg.)*. FAU University Press.

Babbel. (2013, September 24). *So fresh the new iPhone apps are here – with the complete range of courses*. Retrieved January 16 2026, from Babbel Magazine: <https://www.babbel.com/en/magazine/fullversion-iphone-apps>

Babbel GmbH. (n.d.). *Babbel – Sprachen Lernen-app*. Retrieved March 27 2026, from App Store: <https://apps.apple.com/de/app/babbel-sprachen-lernen/id829587759>

Bandhu, D., Mohan, M. M., Nittala, N. A. P., Jadhav, P., Bhaduria, A., & Saxena, K. K. (2024). *Theories of motivation: A comprehensive analysis of human behavior drivers*. *Acta Psychologica*, 244, 104177. <https://doi.org/10.1016/j.actpsy.2024.104177>

Braun, S. (2025, June 9). *Duolingo's CEO outlined his plan to become an "ai-first" company. he didn't expect the human backlash that followed*. Retrieved March 27 2026, from Fortune: <https://fortune.com/2025/06/09/duolingo-ceo-surprised-backlash-ai-first-company-announcement/>

Content-based instruction / content and Language Integrated Learning. Methods of Language Teaching. (n.d.). Retrieved March 27 2026, from Brigham Young University Website: <https://methodsoflanguageteaching.byu.edu/content-based-instruction-content-and-language-integrated-learning>

Bouchrika, I. (2026, March 19). *Competency-based education guide: Benefits differences to traditional education for 2026*. Retrieved March 27 2026, from Research.com: <https://research.com/education/competency-based-education>

Deterding, S., Sicart, M., Nacke, L., O'Hara, K., & Dixon, D. (2011). *Gamification. Using game-design elements in non-gaming contexts*. <https://doi.org/10.1145/1979742.1979575>

Duolingo. (n.d.). *Duolingo – Sprachkurse-App*. Retrieved March 27 2026, from App Store: <https://apps.apple.com/de/app/duolingo-sprachkurse/id570060128>

Duolingo. (2025, April 28). *below is an all-hands email from our CEO, Luis von Ahn*. Retrieved March 27 2026, from LinkedIn: https://www.linkedin.com/posts/duolingo_below-is-an-all-hands-email-from-our-activity-7322560534824865792-l9vh

Duolingo, Inc. (DUOL): History, ownership, Mission, How It Works Makes Money. (n.d.). Retrieved March 27 2026, from dcf-model.com: <https://dcfmodeling.com/blogs/history/duol-history-mission-ownership>

Guldhe, P. (2023, June 27). *Reality is broken book summary*. Retrieved March 27 2026, from Medium: <https://medium.com/@pratiguldhe42/reality-is-broken-book-summary-6af32907cbd6>

Krogue , K. (2012, September 18). *5 gamification rules from the grandfather of gamification*. Retrieved March 27 2026, from Forbes: <https://www.forbes.com/sites/kenkrogue/2012/09/18/5-gamification-rules-from-the-grandfather-of-gamification/>

Jacob. (2023, December 29). *What is the difference between gamification and games? 2024*. Retrieved March 27 2026, from Gamification Hub: <https://www.gamificationhub.org/what-is-the-difference-between-gamification-and-games/>

Jurran, N. (2024, October 15). *Sprachen Lernen: Babel-app Bekommt Ki-Sprechtrainer*. Retrieved March 27 2026, from c't Magazin: <https://www.heise.de/news/Sprachen-lernen-Babel-Appbekommt-KI-Sprechtrainer-9980821.html>

Kurt, S. (2023, September 28). *Gamification, what it is, how it works, examples*. Retrieved March 27 2026, from Educational Technology: <https://educationaltechnology.net/gamification-what-it-is-how-it-works-examples/>

Lewis, B. (2019, November 13). *How to use cooperative learning in the classroom*. Retrieved March 27 2026, from Thought Co.: <https://www.thoughtco.com/what-is-cooperative-learning-2081641>

LingoDeer Co Ltd. (n. d.). *Lingodeer: Sprachen Lernen-app* . Retrieved March 27 2026, from App Store: <https://apps.apple.com/de/app/lingodeer-sprachen-lernen/id1261193709>

Pelling, N. (2011, August 9). *The (short) prehistory of “gamification”... Funding Startups (other impossibilities)*. Retrieved March 27 2026, from: <https://nanodome.wordpress.com/2011/08/09/the-short-prehistory-of-gamification/>

Appendix

Pingo AI Inc. (n.d.). *Ki-Sprachen Lernen: Pingo ai-app*. Retrieved March 27 2026, from App Store: <https://apps.apple.com/de/app/language-learning-pingo-ai/id6738882620>

Pingo AI: Ai Companion that helps you learn languages. (n.d.). Retrieved March 27 2026, from Y Combinator: <https://www.ycombinator.com/companies/pingo-ai>

Professor Sebastian Deterding professor of digital creativity. (n.d.). Retrieved March 27 2026, from University of York: <https://www.york.ac.uk/arts-creative-technologies/people/sebastian-deterding/>

Profile Dr. Zeynep Köylü. (n.d.). Retrieved March 27 2026, from University Basel: <https://english.philhist.unibas.ch/en/persons/zeynep-koeylue/profile/>

Richards, J. C., & Rodgers, T. S. (2001). *Approaches and Methods in Language Teaching* (2. Aufl.). Cambridge University Press. <https://doi.org/10.1017/CBO9780511667305>

Richards, J. C., & Rodgers, T. S. (2022, April 8). The Natural Approach (chapter 14) - approaches and methods in language teaching. Retrieved March 27 2026, from Cambridge Core: <https://www.cambridge.org/core/books/abs/approaches-and-methods-in-language-teaching/natural-approach/9B35DC365BB3A014C5C3D4C08EA140D5#chapter>

Sanchez, E., Van Oostendorp, H., Fijnheer, J. D., & Lavoué, E. (2019). *Gamification*. In A. Tatnall (Hrsg.), *Encyclopedia of Education and Information Technologies* (S. 1–11). Springer International Publishing. https://doi.org/10.1007/978-3-319-60013-0_38-1

Scheidlinger, S. (1948). *A Comparative Study of the Boy Scout Movement in Different National and Social Groups*. American Sociological Review, 13(6), 739. <https://doi.org/10.2307/2086827>

Scott, L. (n.d.). *Lingodeer review: Decent alternative to other language apps*. Retrieved March 27 2026, from The Mezzofanti Guild: <https://www.mezzoguild.com/lingodeer-review/>

Serious games - spielerisch Ernste Inhalte vermitteln. (n.d.). Retrieved March 27 2026, from bildung.digital: <https://www.bildung.digital/artikel/serious-games-spielerisch-ernste-inhalte-vermitteln>

Stangl, W. (n.d.). *Edutainment*. Retrieved March 27 2026, from Online Lexikon fr Psychologie Pädagogik: <https://lexikon.stangl.eu/23405/edutainment>

Sharma, D., & Sharma, J. (2023). *EVOLUTION OF GAMIFICATION, ITS IMPLICATIONS, AND ITS STATISTICAL IMPACT ON THE SOCIETY*. ShodhKosh: Journal of Visual and Performing Arts, 4(2SE). <https://doi.org/10.29121/shodhkosh.v4.i2SE.2023.456>

Sharma, P., & Joshi, K. (2025). *Flow and Its Application in Education*. In G. Dutta, G. Misra, S. Gupta (Hrsg.), *Handbook of Positive Psychology and Indian Thought Systems* (S. 297–303). Springer Nature Singapore. https://doi.org/10.1007/978-981-96-4210-6_19

Shelper, P. (2025, October 23). *The true history of loyalty programs*. Retrieved March 27 2026, from Loyalty Reward Co. <https://loyaltyrewardco.com/the-true-history-of-loyalty-programs/>

Situational language teaching. (2023, June 12). Retrieved March 27 2026, from Teflpedia: https://teflpedia.com/Situational_language_teaching

Trushyna, K. (2025, June 27). *Lingodeer Erfahrungen: Ist Die app Eine Gute Wahl?*. *Online Sprachschule*. Retrieved March 27 2026, from n-tv.de: <https://www.n-tv.de/online-sprachschule/apps/lingodeer/>

Van den Branden, K. (n.d.). *Task-based language teaching: Basic principles and misunderstandings*. Retrieved March 27 2026, from Goethe Institut: <https://www.goethe.de/ins/be/en/spr/mgs/26068586.html>

Wallingford, P. (2024, September 15). *When was duolingo created? The fascinating story*. Retrieved March 27 2026, from Duolingo Guides: <https://duolingoguides.com/when-was-duolingo-created/>

What is brief history of Babbel Company?. Canvas Bussines Model. (2026, March 15). <https://businessmodelcanvastemplate.com/blogs/brief-history/babbel-brief-history>

What are AI hallucinations?. (n. d.). Retrieved March 27 2026, from IBM: <https://www.ibm.com/think/topics/ai-hallucinations>

Xing, M. (2025, April 26). *Pingo AI: Ai Tutor for language learning: Michael Xing posted on the topic*. Retrieved March 27 2026, from LinkedIn: https://www.linkedin.com/posts/michael-xing-04216821a_we-launched-pingo-ai-in-january-by-march-activity-7322003692927229952-uuIC

7.2 Image Bibliography

Image 01

Richards, J. C., & Rodgers, T. S. (2001). *Approaches and Methods in Language Teaching* (2. Aufl.). (p. 8). Cambridge University Press. <https://doi.org/10.1017/CBO9780511667305>

Image 02

Matsuzawa, S. (2019, December 7). *137 merit badges, 137 memories: Eagle Twins Earn Rare Honor of Collecting every boy scout merit badge*. Retrieved March 27 2026, from KTVB: <https://www.ktvb.com/article/news/local/137-merit-badges-137-memories-eagle-twins-earn-rare-honor-of-collecting-every-boy-scout-merit-badge/277-0f1786f5-dba1-4b3e-8444-d51a75b9cbcd>

Image 03

Chavez, C. (2016, March 24). *First official pokemon go screenshots and details*. Retrieved March 27 2026, from Phandroid: <https://phandroid.com/2016/03/24/pokemon-go-game-android-ios/>

Image 04

Sharma, P., & Joshi, K. (2025). *Flow and Its Application in Education*. In G. Dutta, G. Misra, & S. Gupta (Hrsg.), *Handbook of Positive Psychology and Indian Thought Systems* (p. 300). Springer Nature Singapore. https://doi.org/10.1007/978-981-96-4210-6_19

Image 05

Arndt, H. (2022). *Digitale Spiele und fachliches Lernen: Band 1* (H. Arndt, Hrsg.). FAU University Press.

Image 06-09, Image 22-23, Image 25

Duolingo. (n.d.). *Duolingo – Sprachkurse-App*. Retrieved March 27 2026, from App Store: <https://apps.apple.com/de/app/duolingo-sprachkurse/id570060128>

Image 10-13, Image 24

Babbel GmbH. (n.d.). *Babbel – Sprachen Lernen-app*. Retrieved March 27 2026, from App Store: <https://apps.apple.com/de/app/babbel-sprachen-lernen/id829587759>

Image 14-17

LingoDeer Co Ltd. (n. d.). *Lingodeer: Sprachen Lernen-app*. Retrieved March 27 2026, from App Store: <https://apps.apple.com/de/app/lingodeer-sprachen-lernen/id1261193709>

Image 18-21

Pingo AI Inc. (n.d.). *KI-Sprachen Lernen: Pingo AI-App*. Retrieved March 27 2026, from App Store: <https://apps.apple.com/de/app/language-learning-pingo-ai/id6738882620>

7.3 Interview with Dr. Zeynep Köylü, University Basel, 05.03.2026

Which apps do you know exactly, so which apps did you analyze for your works?

Okay, so many years ago I analyzed some learning management systems embedded into some course books from Pearson, Macmillan.

And these were partially interactive LMS systems, I would say. So, it was before, of course, the, you know, generative AI technology, like back in 2010s. That was the first one.

And since then, I haven't investigated any other AI tool, empirically, let's say.

Do you think that gamified learning experiences could help learning a second language?

So, what I found in my own research, working with this language management system, I included in the intervention itself some advanced grammar items, some grammar topics, let's say. And I had three groups of learners. So, one group was only exposed to traditional teacher-fronted learning.

And they did follow practice, again, following like, you know, paper, pen, traditional methods. The second group followed a blended methodology. So, they were both exposed to traditional teacher-fronted learning.

But they also engaged with this learning management system. And in the third group, the group itself only learned those topics over the learning management system. Zero teacher interaction or like teacher introduction, whatever.

And they did only online activities. So, when I compared their performances over a semester, interestingly, of course, in terms of test performance, the teacher-fronted group had the best scores. But in terms of retention, that was so interesting, the online group, like the group only used the learning management system, had the best scores.

So, they outperformed the other two groups. And there were only like negligible difference in terms of the blended group and the

teacher-fronted group in terms of performance. But yeah, I mean, that was really cool.

So, the more cognitively you engage with the material itself, trying to teach, trying to understand and so on, probably you're making more neural connections. Like, I mean, it's more long-lasting, I would say, right? Or maybe that was just random, but the participants in that digital group who only used that language management system were more autonomous learners, I would say. Because all of them had really like good results in terms of retention.

So, I had multiple data collection times to explain to you what retention means. So, they had a class of like, I don't know, the topic for four weeks and then four weeks, an interim test, performance test. Then four more weeks of class, learning another thing, another test.

And of course, they had the pre-test before starting to learn the topic itself. And then when the teaching was done, I think around like six weeks later, they got the retention test. And the best retention rate came from that digital group.

Do you see any risks of learning everything online?

Well, to be able to really learn something digital, like using, exploiting some technology, not necessarily AI or, you know, not necessarily pedagogically developed tools. You can watch a YouTube video and learn some words, right? So, I think what really mediates results here, like favorable results here, would be individual differences.

If you are a learner with high learner autonomy, better self-agency, better motivation, like different working learning strategies and so on, then you will probably end up learning. But you have to be critical about how to consume that material as well. Because sometimes you may reach out to some, you know, like mistaken information, like wrong information, especially in the case of Gen AI, right? I mean, then you have to double check, right? So, you have to have that kind of like responsibility and agency about your own learning.

You have to be critical. If you have that, then any tool might work. But research has shown that we need guidance from a professio-

nal, a teaching professional, right? So, without that guidance, we can easily get lost.

It then becomes just a pastime activity, and you end up learning nothing. Yeah, especially bilingual integrated AI and more. That's rubbish.

Like nobody can learn anything. It's just drills, right? I mean, it's rule-based drill learning. Even if the AI, like giving you feedback, like not something really like worthy of seeing, let's say.

I mean, I don't think that anyone can learn anything with Duolingo. I mean, research has also shown that nobody learned Duolingo. You know, you just study from Leuven et al., Jean Löwen.

Yeah, I mean, and Matthew Kessler in the States, in Michigan State. So, a group of PhD students.

And Matthew Kessler. So, these people had a group of PhD students, graduate students, and they wanted them to learn, self-teach themselves Turkish using Duolingo. Of course, they couldn't end up learning anything.

Yeah, that's just one study that just like comes up to my mind right away. But I mean, I've read quite a few studies about, particularly Duolingo, and the results are like not so satisfying.

Do you see any potential in learning apps for the education system?

I have no idea because I don't know about K-12 right now.

Like in the past, when I was teaching at a university, like English for academic purposes, we had really limited amount of technology integrated into teaching. But now, in Turkey, for example, I can talk about the Swiss system. For example, my daughter is going to third grade, and she is learning French.

And they have this like online management, online learning management system again, LMS again, for that book, Millefeuille, maybe you've heard of it. And they have this like, you know, the interface and so on. So, it's the digitalized version of the book with like actual pronunciation and like, you know, like flashcards and so on with the German translation of the French input and so on. But she never, I mean, she's good with like, I mean, French, her grades and so on, but she doesn't want to play around with that

learning management system. Maybe it's not appealing because of the design. But the teacher keeps assigning some homework from that Mille-feuille thingy, let's say, the online management system. Oh, it is on Mille-feuille.

The teachers love it. Because they don't have to prepare new lessons, you know, because they know it by heart. That might be one reason. New material means new work.

Like you have to work, you have to read the book, you have to prepare new class materials, new worksheets, whatever. And nobody wants to do that. Yeah, that's really disappointing.

I was there and I was like, yeah, that's one thing. I mean, but in Turkey, for example, yeah, there's too much. Like if the child goes to a private institute, then the use of technology is now kind of an indicator of the quality of education offered.

So, these kids are bombarded with many different forms of technology. And I really doubt if it's really useful to them. It's too much screen time, if you ask me.

And it really damages attention spans. So there has to be somewhere in between, probably. But we need, again, guidance, particular frameworks.....

We need expert opinion about that.

Do you see any potential in the education system with LLMs?

So, these language apps or integrated? Yeah, that's difficult. Not for any proficiency level.

So, you have to have this advanced proficiency to be able to tailor towards your own needs. Because what is the selling point with AI technology, for example? Personalization. You can personalize, like there are these AI language tutors, like maybe you know about Call Annie.

Have you ever seen her? That's so crazy. I can show you on my phone. So it's like an avatar, but it's very realistic.

You call her. Just a sec. Yeah, there are many different avatars, many different faces, characters.

The typical one is the American girl Annie. But there are many different languages, many different fantasy characters, animated

characters, whatever, historical figures and so on. So you pick and call that person.

So, you basically call this person, Annie. And just like you engage with a conversation with that person, which requires a certain level of proficiency in the target language. If you're a beginner, it's too difficult, I would say.

And I think the most important risk that we need to take into consideration is that these tools really damage learner languages. Like there is this characteristic learner language, interlanguage. We make mistakes.

We are creative as learners. For example, we use two, let's say, words in a multi-word construction in a weird way because we are learning, right? We do transfer from our first language to other languages that we speak. So, we have this really particular characteristic of learning language.

But AI always, like pumps you with native speakers, perfect native speaker varieties. So, you always see the best example. But is it good? Always, you know, all your mistakes are being corrected right away, but not you correct them.

You see what I mean? So, I think it's not for everyone, for all levels of proficiency. I would say. A lot of them.

They used to have like a community page in Duolingo where the learners are corrected, but they removed it. Now I know. So, it was disappointing.

Now it's AI doing that. So, they removed that community thingy, right? But I think in Busuu, there is still this community.

Do you think it's kind of worrying to learn languages with AI since there's people who become more attached to it?

Yeah, there are. Yeah. So language learning is intrinsically an interactive kind of social act, right? So with this virtual interaction, I don't believe that you can learn a language as you would learn it in social communication.

But some people with neurodiversity or shy characteristics and so on, they sometimes fall into the trap of AI. And yeah, they want to use it all the time. But it's an infatuation kind of like they are abusing the affordances of technology to me, right? But I don't

think that there are too many instances of that.

But it might be a bit isolating and people getting addicted in that sense and preferring AI interaction over human interaction. That's my concern. So we shouldn't be doing so.

Did you observe some kind of psychological mechanism through learning languages and apps or not really?

I have seen some studies looking into the notion of anxiety, for example, and how these tools help you lower down your speaking anxiety. There are some research articles out there about that.

Or how can these tools further motivate you to learn the language. But other than that, I haven't seen. I don't know about literature, about psychology and language learning through AI or Duolingo and so on.

Yeah. So it feels like Duolingo tries to flow the user to feel like they want to... I think the app tries to keep engaged with the user so they would stay. They have a lot of animations.

Yeah. Badges you collect, coins you collect. That's the game.

And it's like... Also, agitation, right? The bird is dying, glitching. The bird is like, hey, why don't you just have a look at me today? A kind of miserable situation the bird gets into in the case of Duolingo. And it's agitation, I think.

Right? The pressure to the user. Yeah. I think it's not okay.

Especially, I think in terms of kids. Like my daughter, she wanted to learn Korean over Duolingo. And we just let her download Duolingo to her iPad.

And yeah, she was like, "but mom..." Normally she has two days over the weekend and limited time for screen consumption. Including either television or any form of screen, okay? And she wanted to use Duolingo within that time frame. But of course, Duolingo sent notifications during times when she's not allowed.

She was not allowed to use any technology or whatever. But she was like, but mom, the bird is dying. I have to take care of the bird. So, I mean, from a kid's perspective, it might be even more of abuse in a sense. Not even oppression. Beyond oppression, I would say. So, we just deleted it. It doesn't have age restrictions. Age restrictions, yeah.

7.4 Use of Aids

Claude.ai was asked for assistance with word choice.
It was explicitly not used as a research aid or to generate writing ideas.
<https://claude.ai/>

Deepl AI Write was used for correcting grammar and spelling.
<https://www.deepl.com/de/write>

TurboScribe was used for making the transscript for the interview with Dr. Zeynep Köylü.
<https://turboscribe.ai/>

Declaration of Originality

I hereby declare,

- that I have correctly cited all text passages that are not my own in accordance with standard academic citation rules and that I have clearly mentioned the sources used;
- that I have declared all tools and AI assistance systems used, chatbots such as ChatGPT, translation and paraphrasing tools, or programming applications such as GitHub Copilot in a directory and indicated their use in the corresponding text passages;
- that I have acquired all immaterial rights to any materials used by me, such as images or graphics, or that these materials were created by me;
- that the topic, the thesis or parts thereof have not already been used in a performance record of another module, unless this has been expressly agreed with the lecturer in advance and is stated in the thesis;
- that I am aware that my work may be checked for plagiarism and third party authorship of human or technical origin (artificial intelligence);
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Basel, 29.03.2026

Place, Date

Signature

A handwritten signature in black ink, appearing to read 'Serem', is written over a horizontal line.

